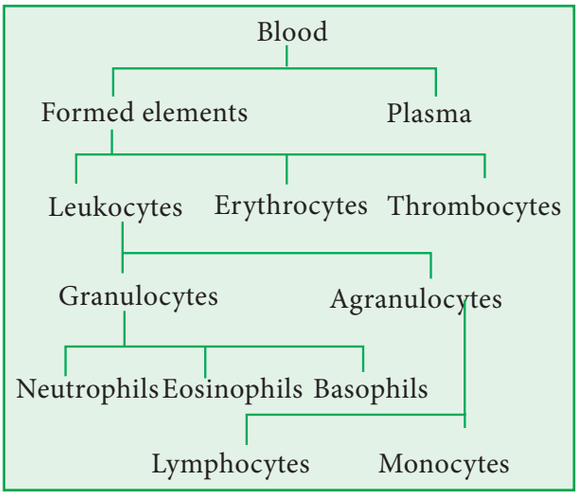
proteins and most of the formed elements are left in the blood vessels. This fluid is called the interstitial fluid or tissue fluid. Exchange of nutrients, gases, etc. between the blood and the cells always occur through this fluid. There is an elaborate network of vessels called the lymphatic system. The lymphatic system collects this lymph fluid and drains it back to the major veins such as thoracic duct and subclavian vein.

Fats are absorbed through lymph in the lacteals present in the intestinal villi. Lymph drains fluid from extracellular or intercellular spaces into blood. It is used to maintain a balance between blood and interstitial fluid.

Blood

Blood functions as a vehicle for mass transport of materials between cells and the environment or between the cells themselves for maintaining homeostasis. Blood consists of cellular portion called as formed elements (cells) that are suspended and carried in a fluid portion called plasma. The total blood volume in an adult is about 5 litres. The normal pH range of the blood is 7.35 – 7.40.

When a blood sample is centrifuged, the heavier formed elements are packed at the bottom of the centrifuge tube, leaving plasma at the top. The formed elements constitute 45% of total blood volume and the plasma accounts for the remaining 55%.



The formed elements include erythrocytes, leukocytes and platelets (thrombocytes). A cubic millimeter of adult blood normally contains 4.9 million to 5.5 million erythrocytes in males and 4.4 million to 5.0 million erythrocytes in females. The number of leukocytes in an adult human being is 5000 to 9000 leukocytes per cubic millimeter. Leukocytes include granulocytes (neutrophils, eosinophils, basophils) and agranulocytes (lymphocytes and monocytes). The normal platelet count in the blood is between 150000 and 300000 cells per cubic millimeter.

Plasma is a pale-yellow coloured liquid consisting of water and dissolved solutes. The solutes include ions like Na^+ as well as organic molecules such as metabolites, hormones, enzymes, albumins, globulins, fibrinogen and other proteins.

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